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The Editor
Quantum Science and Technology
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Dear Editor,

I am pleased to submit the manuscript “Capacity, Not Barren Plateaus: Diagnosing Variational Quantum Training Failure on Chaotic Dynamics, and a Fixed-Reservoir Resolution” for consideration as a research paper in *Quantum Science and Technology*. The work is original, has not been published elsewhere, and is not under consideration at any other journal.

The paper makes a diagnostic contribution that I believe is essential reading for the quantum-machine-learning sub-field. Variational quantum-circuit failures on hard learning tasks at NISQ scale are routinely attributed to barren plateaus — exponentially vanishing gradients in deep parameterised circuits — and a substantial volume of recent work proposes mitigations targeted at this specific failure mode. We demonstrate, on a benchmark chaotic-dynamics task at the relevant hardware scale, that this attribution is empirically and theoretically wrong. The McClean barren-plateau bound for the relevant ansatz class predicts gradient magnitudes that are four orders of magnitude below those we measure. The dominant failure mode is capacity-limited expressivity rather than gradient vanishing, and mitigations targeted at the gradient-vanishing regime should not be expected to help.

This diagnostic finding has lasting consequences. First, it constrains which mitigation strategies are appropriate at NISQ scale, redirecting effort toward architectural rather than gradient-side interventions. Second, the diagnostic methodology — the direct comparison of measured gradient norms against the McClean threshold for the ansatz class — generalises to any variational quantum-machine-learning study and should be standard practice. Third, the finding clarifies what an architectural alternative needs to bypass: not gradients per se, but the underlying capacity limit. We illustrate this with a fully specified Quantum Reservoir Computing pipeline that achieves 81% lower mean-squared error and trains approximately $52,000\times$ faster on matched quantum resources, with the same architecture generalising across three canonical chaotic systems without per-system retuning.

The paper additionally contributes an empirical scaling sweep on Lorenz, $q \in \{5, \dots, 10\}$ against the standard ESN baseline at $N_{\text{ESN}} = 500$, showing that the naive feature-dimension parity crossover does not occur: QRC test MSE degrades monotonically with qubit count at fixed prediction-target dimension because features dilute relative to a fixed-dimensional signal. The correct scaling axis is matched-complexity, in which qubit count and target dimension grow together; this is the subject of a separate companion manuscript in preparation. The QST paper also contributes a windowed-variational quantum-circuit (WVQC) ablation that closes the information-asymmetry concern in the QRC–QPINN comparison. The WVQC ablation, in which a variational circuit is given the same windowed input as QRC and trained by Adam under the same protocol as QPINN, recovers only approximately 32% of the QRC–QPINN gap on matched seeds, isolating the fixed-reservoir architectural choice as the dominant factor and extending the capacity-limit diagnosis to the information-symmetric setting.

All experimental code, data, and figures are released publicly at the repository linked in the manuscript, supporting reproducibility and extension by other groups.

I have intentionally restructured the paper so that the diagnostic finding leads, with the architectural comparison serving as supporting evidence rather than as the primary claim. I believe this ordering reflects the lasting contribution and is appropriate for the journal's selectivity criterion.

I would be grateful for your consideration. I am happy to suggest reviewers if helpful.

Sincerely,

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